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RESEARCH

Robust audio watermarking using adaptive embedding in intrinsic modes

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Abstract
This paper presents a robust and blind audio watermarking technique utilizing iterative filtering for adaptively embedding encrypted watermark data into perceptually insignificant regions of the audio signal. This work aims to address the growing challenge of intellectual property (IP) infringement and fill the existing research gap by developing a watermarking method that guarantees audio quality alongside strong resistance to illegal attacks. The objective of this research is to design a watermarking scheme that can jointly ensure robustness, imperceptibility, security, and blind extraction, specifically under advanced signal processing, desynchronization, and Stirmark attacks. The core of the proposed scheme is iterative filtering, which decomposes the host audio signal into various intrinsic mode functions (IMFs) and a low-frequency residue component. The final residue, being crucial for maintaining perceptual quality, is selected for watermark embedding. The residue is segmented into 8×8 non-overlapping blocks, and an adaptive threshold is computed for each block based on its statistical features. Watermark embedding is carried out by modifying specific samples using a novel adaptive strategy, wherein the selected sample is adjusted to a quarter fraction of the product of the local threshold and an embedding strength factor, depending on the watermark bit value. Chaotic and DNA encryption is employed to guarantee the security of watermark. Experimental results confirm high imperceptibility (SNR: 37–40 dB), perfect recovery (NCC = 1), and robust performance with a payload of 204.8 bps.

Keywords Adaptive audio watermarking · Copyright protection · Robustness · Imperceptibility · Iterative filtering · Desynchronization

1 Introduction

The advent of digitization in contemporary times has resulted in significant transformations across the globe. The third decade of the twenty-first century has witnessed a burgeoning influx of multimedia, with the free flow of information predominantly taking the form of various media formats. Information has become a paramount resource for human society in the modern era. However, the proliferation of computer networks has made the exchange of digital files increasingly susceptible to unauthorized access, intrusions, and breaches, (Zhao et al., 2021).

Numerous informative, educational, and social websites, widely used worldwide, have fallen victim to illegal access, prompting a surge in the issue of copyright infringement. In response, there is a rising interest in research in this domain. Watermarking via the digital domain emerges as a potentially useful strategy to combat copyright infringement, offering the capability to verify digital files and uphold the rights of content owners (A. K. Sahu, 2024). While extensive research has been conducted on image watermarking applications (Mehraj et al., 2022, 2023), the demand for effective frameworks for embedding watermarks in audio files has intensified owing to continua issues such as music piracy. According to a 2024 report from the Recording Industry Association of America (RIAA), music piracy costs the industry an estimated $12.5 billion annually through illicit downloading (IPI, 2024). Additionally, a U.K.-based Technology Company named MUSO, specializing in anti-piracy

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solutions and digital rights management services, recorded over 17 billion visits to music piracy sites in 2024 (WIRED, 2024; MUSO, 2024). Furthermore, the widespread use of audio editing programs like Adobe Audition and Cool Edit Pro has enabled individuals to swiftly create and deploy original audio files, including audiobooks, aural effects, and songs. This has raised worries among the contributors about the illegitimate use of their primary audio materials on the internet (Zhang et al., 2023). In response to such challenges, one of the widely adopted solutions is audio watermarking, which involves embedding distinct text, sounds, or images into a source audio file prior to exporting, thereby making it recognizable to copyright owners. This serves as a deterrent for illegal users attempting to misappropriate the original audio materials.

Most popular audio watermarking techniques available today can be divided into two categories: time domain (A. K. Sahu, 2021) and transform domain techniques. Watermarking in the time domain entails directly injecting the watermark bits into the audio stream by altering the loudness of each audio sample. Conversely, watermark bits are incorporated into the transform coefficients of host audio samples in the transform domain. Studies have indicated a stronger preference for frequency-domain methods over time-based ones because of their improved robustness and perceptual quality (Hua et al., 2016).

The demand for robust watermarking has increased, considering the array of attacks that go beyond common signal processing threats, such as desynchronization and stirmark attacks, impacting the seamless transmission of audio data. Desynchronization attacks deliberately disrupt the synchronization or temporal alignment of various audio channels or components. These attacks aim to shift or misalign the timing of audio signals, leading to coherence loss, perceptual changes, or diminished audio quality. Conversely, stirmark attacks serve as benchmark challenges to assess the resilience of audio watermarking algorithms under pressure (Zhao et al., 2021). This evaluation focuses on the watermark's resistance to signal processing operations altering the frequency content of the audio data. Numerous robust audio watermarking techniques in the literature are based on the frequency domain, including the discrete Fourier transformation (DFT) domain (Hu et al., 2021), the discrete wavelet transformation (DWT) domain (Singha & Ullah, 2022), and Fast Fourier transform (Hu et al., 2019) the discrete cosine transformation (DCT) domain (Liu et al., 2021). Each of these approaches has been developed to assess robustness against various types of attacks. For instance, the scheme presented in (Hu et al., 2021) demonstrates resilience against common signal-processing attacks and some desynchronization attacks. However, it has not undergone evaluation for stirmark attacks.

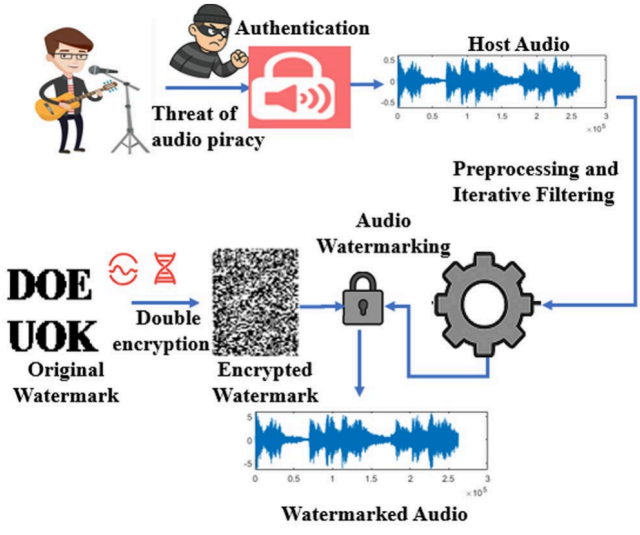


Fig. 1 Visual representation of the present watermarking system

The technique presented in (Singha et al., 2021) focused on a limited set of signal-processing attacks, while schemes in (Lui et al., 2021; Zhao, 2023) and (Lui et al., 2019) underwent analysis solely for common signal-processing and desynchronization attacks. Notably, none of these schemes have been evaluated for stirmark attacks. There is a recognized gap in the literature, as only a handful of algorithms have been scrutinized for stirmark attacks, indicating a need for further improvement to enhance their robustness against these specific types of attacks.

Given the imperative for digital data security and privacy, this work aims to establish a secure and robust framework for the safeguarding of copyright and content verification of music and speech data. The framework aims to demonstrate resilience not only against common signal processing and desynchronization attacks but also considers stirmark attacks. The primary objective is to enhance the privacy and security of audio transmissions while maintaining imperceptibility, capacity, and robustness. To achieve this goal, this work introduces a new method of blind, secure, and resilient watermarking of audio signals utilizing iterative filtering. Iterative filtering is a data-driven strategy that decomposes audio signals into intrinsic mode functions (IMFs) and a residue component, without requiring predefined basis functions like Fourier, wavelets, etc. This makes it adaptable and well-suited for audio signals. The watermark embedded in the residue IMF using iterative filtering gives excellent robustness against different types of attacks and, at the same time maintains a high imperceptibility and payload capacity (Naqash et al., 2023). For a clearer illustration, we have presented a visual representation of the entire scenario in Fig. 1, highlighting the overall pictorial representation of the present watermarking system.

The main contributions of this paper are:

- This research introduces a robust iterative filtering technique for watermark embedding that maintains audio quality while resisting various attacks.
- The proposed robust audio watermarking system functions completely in blind manner; data extraction does not require the host audio or the original watermark.
- Compared to state-of-the-art methodologies, our study shows higher resistance against numerous sorts of attacks, assuring robustness against desynchronization and stirmark attacks as well apart from common signal processing attacks. For audio data, this robustness ensures efficient copyright protection.

The present paper is organized as follows: Section 2 presents a literature survey of the existing literature; Section 3 presents the mathematical preliminaries; Section 4 presents a suggested data-driven robust audio watermark framework; Section 5 presents the experimental results; and Section 6 presents the paper's conclusion.

2 Related work

Among the numerous methods for securing digital multimedia systems, digital watermarking has proven to be a successful strategy (Wu et al., 2020; Xu et al., 2021; Kaur et al., 2017; Lui et al., 2019). Various audio watermarking approaches, either in the spatial or transform domain, have been presented in various works for content authentication and IPR protection. By employing graph-based transform (GBT), DWT, and SVD to generate transformation parameters from the host audio signal, the authors in (Zhang et al., 2023) presented a spread spectrum (SS) based watermarking architecture. To increase the watermark's security, the bits that are to be embedded are first sent to a chaotic encryption algorithm before being inserted into the host audio's coefficients. Although the strategy becomes a bit complex, it provides a stronger defense against many types of attacks, especially cropping attacks. By retrieving the segmental singular values summation (SSVS) and segmental singular values difference (SSVD) characteristic of the original data, a DCT and SVD-based robust audio watermarking approach has been proposed in (Zhao, 2023).

Numerous typical signal processing and desynchronization attacks have been tested against the method. In (Lui et al., 2019), a robust framework for audio watermarking is introduced. First, the frequency-domain-based derivatives logarithmic mean characteristic of the audio sample is calculated, and then a patchwork technique is used to incorporate the watermark. The technique becomes more resilient to several forms of desynchronization and recapturing attacks. A resilient audio watermarking strategy relying on DCT and frequency Singular value coefficient (FSVC) has been proposed in (Hu et al., 2015). In this strategy, DCT is applied to two heeding fragments of the host sample, and then SVD is adopted to each fragment's middle-frequency band DCT coefficients to take out the FSVC. The pixels of the watermark are then implanted by modifying the FSVC values. Better robustness is achieved by the scheme. Nonetheless, there is still room to improve the technique's imperceptibility. In (Lu et al., 2020), a stationary wavelet transform (SWT) based robust audio watermarking scheme based on SS is proposed. To extract feature points, the authors computed the second-order derivative's largest response value from the original audio signal. Although the method offers more defense against cropping and pitch irreversible Time-Scale Modification (TSM) attacks, but, the scheme is less imperceptible. To facilitate the watermark insertion and detection operations, (Charfeddine et al., 2022) created an audio watermarking approach employing DCT and a neural network architecture. By embedding the watermark in the restricted middle-frequency region of an audio frame, the method achieves robustness and strong security while maintaining a high watermark capacity. To add a watermark to the original audio signal, the authors of (Salah et al., 2021) proposed an audio watermarking framework using three different Fourier transforms: DFT, a fractional Fourier transform, and a quaternion discrete Fourier transform. The results obtained after the watermark is embedded for each transform show that the suggested method provides superior resilience against different types of attacks. Additional improvements are required to enhance the technique's resilience and imperceptibility. The fast Fourier transformation (FFT) domain is utilized by two modules in (Aabdelwahab et al., 2020) to give a dependable audio watermarking solution. The first module, referred to as downward progressive quantization index modulation, uses principles deduced from human auditory masking features to modulate the vector norms produced from FFT coefficients. The final voice waveform is produced with a smooth transition between frames thanks to the second module, boundary-constrained iterative correction. Only a few attacks have been tested with the strategy. A digital finite impulse response (FIR) filter design for a time-spread echo-based watermarking technique was reported in (Hua et al., 2015) The design is specifically implemented using convex optimization and spectral factorization techniques. The suggested strategy is primarily concerned with increasing imperceptibility. A new, reliable, and blind audio watermarking method has been created in (Yamni et al., 2022) by combining the fractional Charlier moment transformation (FCMT) and the dual-tree complex wavelet transformation (DTCWT). Every segment of the original audio signal was subjected to the three-level DTCWT utilizing filters specifically designed to

have shift-invariance. The DTCWT estimated coefficients were then subjected to FCMT. Quantized watermark bits have been added to the relative energies of the FCMT coefficients. The algorithm can yet be strengthened by making it more robust to different attacks. A robust audio watermarking scheme for secure transmission of medical images in Medical Internet of Things (MIoT) applications was proposed in (Alshathri et al., 2023) The method employed Wavelet-based Image Fusion, Arnold transforms, and SVD to enhance payload capacity and security. The imperceptibility of the scheme can be further enhanced.

Some prominent issues in the discussed literature that need to be addressed are summarized as follows:

- The literature contains many techniques, such as in (Xu et al., 2021); (Salah et al., 2021); and (Yamni et al., 2022) that perform better against common signal processing attacks but are less resilient against various desynchronization attacks.
- The capacity of the majority of schemes, such as (Zhao, 2023); and (Hu et al., 2023) (Hu et al., 2019) are minimal.
- Few strategies, including those found in (Wu et al., 2020; Zhao et al., 2021, 2023) have been studied for desynchronization and stirmark attacks, and their effectiveness is lacking.

Additionally, Table 1 presents a summary of the pros and cons of the available literature, highlighting key strengths and limitations of each study.

The proposed audio watermarking scheme is quite robust, accounting for all of the aforementioned problems, and has the potential to be very successful in preventing audio data copyright infringement.

3 Mathematical preliminary

This section presents the basic mathematical formulations and theoretical backgrounds that underpin the proposed watermarking framework and is explained as follows:

3.1 Iterative filtering

This section briefly introduces the initial steps for our proposed solely data-driven method-based audio watermarking utilizing an iterative filtering technique. Relying only on the properties of the data itself, rather than predetermined basis functions or transformation techniques like FFT or DWT, Iterative Filtering adaptively decomposes a signal into Intrinsic Mode Functions (IMFs). This makes it a data-driven approach (Naqash et al., 2023).

Table 1 Pros and cons of the available literature

	Transform Used	Pros	Cons
(Zhao, 2023)	DCT and SVD	• Robust audio watermarking scheme has been presented by extracting the SSVS and SSVD features of host data. • The scheme has been tested for the standard signal processing and desynchronization attacks.	• The embedding capacity of the scheme is very low. • The imperceptibility can be enhanced further.
(Zhang et al., 2023)	Graph-based transform (GBT), DWT, and SVD	• A robust spread spectrum (SS) based audio watermarking framework has been proposed. • The chaotic encryption has been performed for enhancing the security of the watermark.	• None of the stirmark attack has been performed on the scheme. • The embedding capacity can be enhanced further.
(Hu et al., 2023)	Fast Fourier Transform (FFT)	• Dual-mode embedding (binary logos and Color images) is performed. • The scheme attains of payload of 508.85 bps. • Achieves strong robustness against signal processing, desynchronization, and compression attacks.	• The computationally complexity of the given scheme has not been explicitly stated. • The imperceptibility of the given scheme is low.
(Hwang et al., 2018)	DWPT, SVD, and QIM	• The use of auditory masking thresholds and adaptive QIM provides strong embedding strength while preserving the host audio. • The scheme attains better robustness for various types of attacks.	• The perceptual quality and payload capacity of the given scheme can be enhanced further. • The use of multiple transformations makes system complex.
(Naqash et al., 2023)	SVD & Iterative filtering (IF)	• Transform domain watermarking approach has been presented, using data driven approach. • By exploiting the IF, finer resolution and better frequency decomposition has been achieved thereby enhancing robustness and sensitivity.	• The scheme has been evaluated for few attacks only. • The scheme lacks evaluation for desynchronization and stirmark attacks
(Yamni et al., 2022)	fractional Charlier moment transform (FrCMT)	• Watermark embedding is achieved employing dual tree complex wavelet transform and FrCMT. • Better robustness is achieved after testing the scheme for common signal processing attacks.	• The scheme has not been evaluated for desynchronization and stirmark attack scenarios. • The computational complexity has not been evaluated.

The conventional empirical mode decomposition (EMD) breaks down a non-stationary signal into a group of the band-restricted functions in the time–frequency domain known as intrinsic mode functions (IMFs) using a recursive process called sifting. The mathematical expression for each IMF is

$$n_l\left(t\right) = b_l\left(t\right) cos\mu_l\left(t\right), \tag{1}$$

Here $b_l\left(t\right), \mu_l > 0 \forall (t)$

It is such that the difference between the number of extrema and zero crossings is not greater than one, and the mean of the upper and lower envelopes added together is zero (Naqash et al., 2023). The standard method relies on an exceedingly unstable filtering procedure and uses haphazard cubic spline proximity to estimate the borders from the maximum and minimum values, which is one of its basic problems because it lacks a good mathematical theory. This makes the operation very sensitive to variations in the input signal. Due to the sifting operation's extreme instability and nonlinearity, convergence is likewise not guaranteed. To avoid computing the average for the sifting procedure, researchers developed a novel method termed iterative filtering, which creates a dynamic moving average operation that is data-dependent instead of finding the envelopes' averages derived using peaks. A flexible local weighted mean is selected to guarantee convergence of the sifting process for simplicity of use.

The sifting can be described as follows by applying a moving mean operator ω on data X.

$$\Upsilon = \mathrm{X} - \omega\left(\mathrm{X}\right) \tag{2}$$

After applying operator ω on data X by j times in a row, we get the first IMF as

$$\delta_1 = lim_{j\to\infty}\Upsilon^j\left(\mathrm{X}\right) \tag{3}$$

The *k*th IMF is often computed as

$$\delta_k = lim_{j\to\infty}\Upsilon^j\left(\mathrm{X} - \delta_1 - \delta_2 - \delta_3 - \ldots . \delta_k - 1\right) \tag{4}$$

The filtering process ends when there is just one local maximum or the local minimum in the data set

$$D = \mathrm{X} - \delta_1 - \delta_2 - \delta_3 - \ldots . \delta_k - 1$$

.

Adaptively selected uniform masks for the operator ' ω' are utilized by the iterative filters to extract the IMFs. Such a window's dimensions can be selected as

$$n = \beta\frac{N}{l} \tag{5}$$

The optimal results are obtained when β is changed between Eq. 2 and Eq. 3. In this case, N denotes the data's size and l denotes the input data's peak count. This approach is comparable to the traditional EMD method in that iterations are terminated according to a standard deviation-based criterion. The number of IMFs obtained depends on the selected SD value, which is usually between 0.001 and 0.2. More IMFs are produced by a smaller SD and vice versa. When the residue signal reaches a certain threshold or turns monotone, demonstrating that no more IMF derivation is possible, the operation stops. It is possible to recover the original signal s(t) from the created IMFs by treating the final residue c_d as an IMF.

$$\mathrm{s}\left(\mathrm{t}\right) = G_i\left(t\right) + \mathrm{c_d}\left(1 \leq \ i \leq \mathrm{d} - 1\right) \tag{6}$$

Figure 2 depicts the IMF generated from the audio signal employed for the present watermarking system. These functions are the representation of the various frequency constituents in the audio signal, ordered in frequency from highest to lowest. The figure also illustrates the residue, commonly known as the audio file's lowest frequency component, providing a comprehensive overview of the frequency distribution. This low-frequency component called residue is utilized here for embedding the watermark to ensure better imperceptibility as well as great robustness against a diverse set of attacks including common signal processing, desynchronization, and stirmark attacks. Due to the utilization of the least frequency constituent (i.e., residue part), the proposed watermarking system achieves optimal results in balancing the trade-off among imperceptibility, robustness, and payload capacity.

3.2 Adaptive embedding

In this paper, we present a new concept of iterative filtering-based adaptive embedding technique where the host signal is first decomposed employing iterative filtering, and the residue component i.e., the last IMF is selected and is manipulated using block-based adaptive thresholding to embed the watermark securely within the audio stream. Adaptive embedding, as implemented in the proposed work, operates by leveraging statistical properties of the audio signal and can be mathematically described as follows:

Let P0 be the original host signal. We perform iterative filtering to obtain the IMF's and the residue component R as follows:

$$\left(\mathrm{IMF}, \mathrm{R}\right) = \mathrm{IF}\left(\mathrm{P0}\right) \tag{7}$$

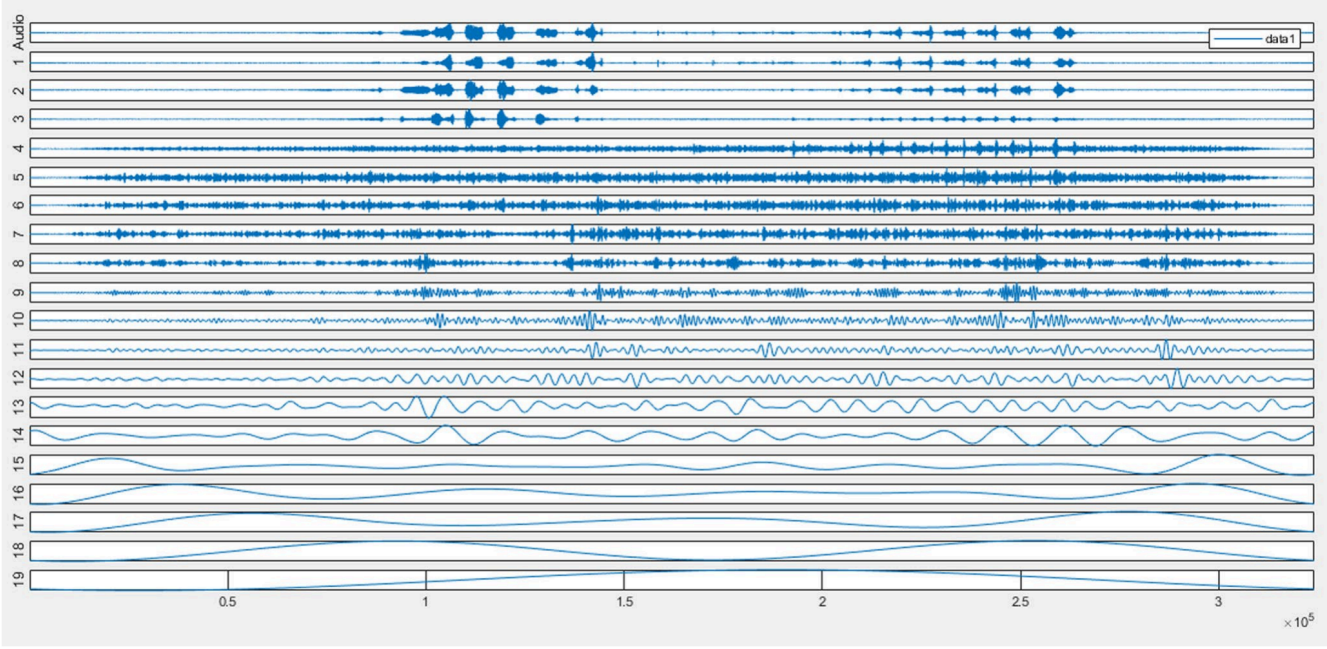


Fig. 2 IMFs of an audio signal

where IMF represents the intrinsic mode functions and R is the residue.

The residue part (R) is reshaped from vector to matrix form as follows:

$$\mathrm{R}\,(\mathrm{q} \times \mathrm{q}) \leftarrow \mathrm{R}\,(1 \times \mathrm{p}) \tag{8}$$

where p represents the original signal length and q × q is the reshaped matrix dimensions.

Perform 8 × 8 non-overlapping blocking of R
For every block, the adaptive threshold represented by η, is calculated based on the block statistics:

$$\eta = (\mathrm{sum}\,(\mathrm{sum}\,(\mathrm{bu}\,(\mathrm{i,j})))) / (\mathrm{a} \times \mathrm{b}) \tag{9}$$

where a and b represent the block dimensions.

Afterwards, the adaptive embedding in each block is performed using the corresponding watermark bit (Wb) as per the equations:

$$\mathrm{bu}\,(\mathrm{i,j}) = \max[1/4\,\lfloor \eta\,|\beta 2| \rfloor\,,\ \mathrm{bu}\,(\mathrm{i,j})] \quad \text{for Wb} = 1 \tag{10}$$

$$\mathrm{bu}\,(\mathrm{i,j}) = \min\,[-\,|1/4\,\lfloor \eta\,|\beta 2| \rfloor|\,,\ \mathrm{bu}\,(\mathrm{i,j})] \quad \text{for Wb} = 0 \tag{11}$$

where bu(i,j) is the preselected sample of the block, β is the embedding strength, ⌊.⌋ represents the floor function, and max[.] and min[.] return the maximum and minimum values of their arguments respectively.

This process is repeated for each block, thereby embedding the entire watermark throughout the residue component of the signal. This technique uses an adaptive embedding strategy to ensure that the watermark is more robustly embedded, yet maintains the perceptual features of the audio signal. The use of an adaptive embedding strategy ensures a striking balance between imperceptibility and robustness

4 The proposed scheme

The proposed bind, secure, and robust audio watermarking technique employs the data-driven approach for embedding the watermark. The data-driven approach employed here is an iterative filtering methodology, which breaks down the audio signal into two parts: a residue component, which is the final component, and a band of IMFs. Since this residue includes low-frequency audio components, it is perfect for the watermark embedding. Since it is a low-frequency component, it barely and minutely alters the audio signal's listening experience, so, making it a perfect applicant for the watermark insertion and extraction process. The block diagram of the proposed framework has been shown in Fig. 3.

The proposed framework uses the host audio's.*wav* file type as it is an uncompressed file format that preserves the standard of the audio while incorporating the watermark. The suggested technique extracts significant features from host

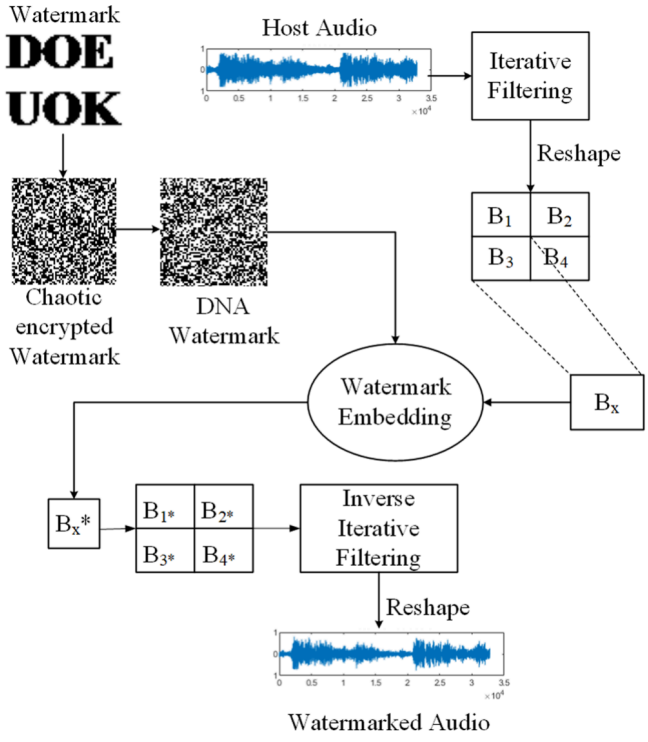


Fig. 3 Block diagram of the proposed audio watermarking framework

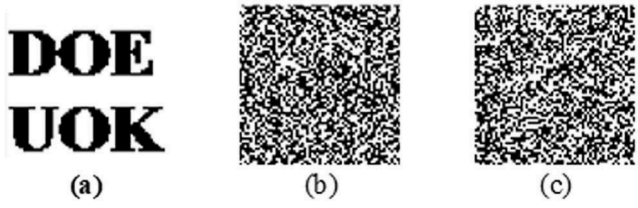


Fig. 4 (**a**) Original watermark (**b**) Ist level encryption (**c**) 2nd level encryption

audio data, leading to improved imperceptibility, embedding capacity, and great robustness. The block schematic of the suggested audio watermarking framework is shown in Fig. 4. Furthermore, the security of watermarks has been strengthened by the combination of chaotic encryption and DNA encryption. Both the initial shape of the watermark and its 2-level encoded version following encryption are shown in Fig. 4. "Algorithm A" explains how to embed the watermark in the suggested architecture, and the opposite of the same algorithm is continued to retrieve the watermark at the end.

This innovative, efficient, and authentication-based robust audio watermarking technique consists of below-mentioned steps: a) Preparation of Watermark, b) Embedding of Watermark, and c) Extraction of Watermark. The three steps are explained in full in the sections that follow:

4.1 Preparation of watermark

Any unapproved user who can figure out the watermark embedding algorithm could jeopardize the security of the system. Two encryption approaches that we have used to increase the security of the recommended technique are DNA-based and chaotic-based encryption, to solve this problem.

We used a chaotic key sequence generated by the logistic map for the first encryption stage. When a logistic map is created, an aperiodic pattern of chaotic values {Nr} is produced. These numbers, which range from 0 to 1, are incredibly unpredictable. Numerical expression for this would be:

$$N_{r+1} = dN_r\left(1 - N_r\right) \tag{12}$$

The bifurcation parameter (d) lies between 0–4. The initial kind of encryption process is established by combining the bits of watermark and the final sequence using XOR. Despite being computationally advantageous, a one-dimensional logistic map with chaotic encryption offers low security because of its small key size, which is vulnerable to brute-force attacks. The already encrypted watermark is encrypted once more using DNA-based encryption to get around these problems. DNA molecules, which encompass four nucleotides A (adenine), G (guanine), T (thymine), and C (cytosine), maintain information for use in computations using DNA, where G and C as well as A and T complement each other (Mehraj et al., 2023), in binary codes, 00 and 11, 10 and 01 complement to one another, so, these bases can be represented by using such arrangements. As claimed by Watson and Crick, only 8 of the 24 feasible pairings have complementary norms. This nucleotide pattern will serve as the key for XORing the DNA sequence to produce a dual-encoded watermark. The positive aspect of encryption based on DNA is that there's very little chance of finding the buried data because between 16 crore and 30 lakh DNA sequences are already accessible. An unapproved user must also be able to decipher the DNA sequence and the contrasting technique to uncover the integrated hidden message. Moreover, the security of the watermark can be validated through the visual difference between the original and dual encrypted watermark, as visualized in Fig. 4. The watermark encryption is performed using a combination of chaotic maps and DNA encoding algorithms. These algorithms provide a very large key space, making brute-force attacks highly unlikely. The encryption is also highly sensitive to small changes in the key even a minute variation results in a completely different encrypted watermark. In addition to this, the encrypted watermark appears visually random and uncorrelated with the original, which enhances

its confidentiality and makes it resistant to unauthorized detection, interpretation, or removal.

4.2 Embedding of watermark

The proposed framework utilizes a data-driven based iterative filtering approach for inserting a resilient watermark. The host audio signal (1 × p) is first subjected to iterative filtering that divides it into a spectrum of IMFs and a residue constituent that remains at the end. This approach does not change the size of the host signal i.e., each IMF as residue will have the size as that of the original host data. The residue component (1 × p) is selected and is converted into (q × q), which is followed by 8 × 8 non-overlapping blocking. Before embedding the watermark, the embedding parameters η is generated from each block that represents threshold reference value, and the watermarking is done according to the procedure mentioned in 'Algorithm A'

Algorithm A Procedure for Watermark Embedding

Input: Original host signal (P0), Watermark (W)

1) ***W=logo1 (n×n)***
2) ***Wc=chaotic encryption (W)***
3) ***We=DNA encryption (Wc)***
4) ***P0=host audio (1×p)***
5) ***Perform iterative filtering of P0***
   [IMF & R] = IF(P0)
6) ***Perform reshaping of R to obtain***
   R(q×q)← R(1×p)
7) ***Perform 8×8 blocking of R(q×q)***
8) ***Select bu (i,j) from all of the blocks of R, and a threshold quantity (η)***
9) ***Use the below mentioned formula to insert the watermark***
   for We=1

$$bu\,(i,j) = max\,[\tfrac{1}{4}|\eta|\beta^2||, \mathrm{bu}(i,j)\,]$$

Otherwise

$$\mathrm{bu}\,(i,j) = min[\,-|\frac{1}{4}|\eta|\beta^2||, \mathrm{bu}\,(i,j)\,]$$

where $\eta = \frac{sum(sum(\mathrm{bu}\,(i,j)))}{a*b}$, ***a and b represents size of the block.***

$\beta = 0.03g$ ***is the embedding parameter***

10) ***Afterward, convert R(q×q) back into R (1×p) and perform inverse iterative filtering to obtain the final watermarked audio***

The function arguments' maximum and minimum are returned by max [.] and min [.], respectively.

Output: Watermarked Audio (Aw)

4.3 Extraction of watermark

To acquire the watermark information back, the watermarked audio (Aw) is selected and is is subjected to iterative filtering, followed by the selection of residue component (q × q) which is subsequently divided into 8 × 8 non-overlapping blocks. On pre-selected pixels, a reverse embedding method is used to extract the watermark bits.

5 Experimental results

Our approach was experimentally tested using MATLAB software version 2014a and an Intel (R) Core (TM) i7-7700 CPU running at 3.60 GHz running Windows OS. Within this segment, we assess the effectiveness of the suggested audio watermarking method both theoretically and experimentally from three perspectives: the embedding capacity (payload) in terms of bps, the robustness against attacks in terms of NCC and BER, and the perceptual quality of the watermarked signals in terms of SNR. For experimental purposes, 60 audio loops have been assessed in this study (10 samples from each of the following categories: classic, pop, rock, jazz, folk, and hip hop). Each audio clip has been taken from Looperman website and lasts 20 seconds having a sixteen-bit resolution at a sampling rate of 44.1 kHz. The watermark has a 64 by 64-pixel dimension. The ratio of the entire quantity of watermark bits embedded (4096 bits) to the total time of the host audio signal (20 sec.) is used to determine the embedding capacity. As a result, the suggested watermarking system has an embedding capacity of 204.8 bit/s (Naqash et al., 2023). Although the original audio tracks were two-channel, we added the watermark using the left channel. We evaluated the proposed method in terms of robustness, audibility, and security. Figure 5 displays a few host audio files used in this work, with the X- and Y-directions representing audio samples and amplitude, respectively. Various quality-assessing measures, such as Bit Error Rate (BER), Normalized Correlation Coefficient (NCC), and Signal-to-Noise Ratio (SNR) are employed for the analysis of the proposed framework.

5.1 Imperceptibility analysis

When analyzing an audio signal's imperceptibility, one typically looks at its characteristics to see if its waveforms can be distinguished from one another. In each case of the given algorithm, the SNR obtained is more than 37 dB under no attack, indicating the excellent transparency of the suggested method. Figure 6 displays the fidelity of a couple of the data-embedded audio files along with retrieved watermarks under no attack. For a variety of audio sample types

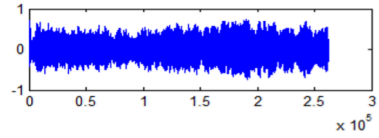
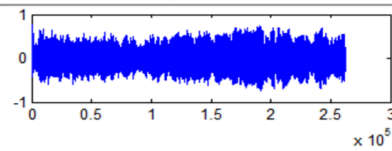
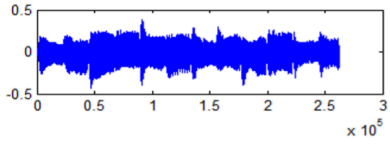
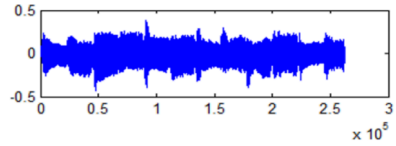
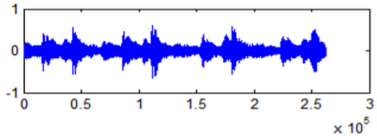
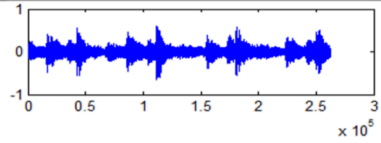
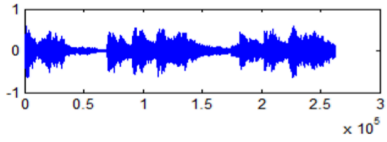
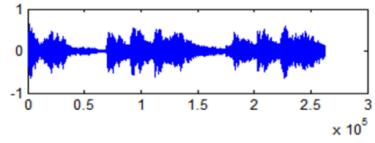
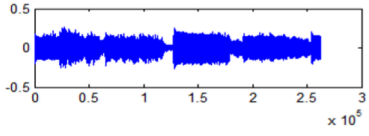
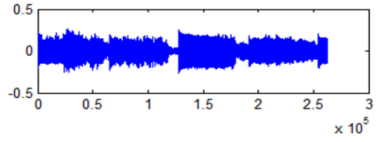
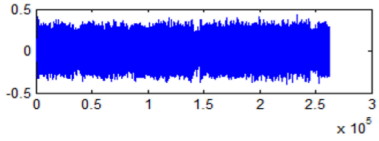
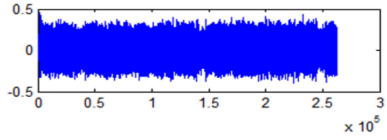
Host audio	Watermarked audio	SNR (dB)
Audio type: classic		
		40.0065
Audio type: folk		
		39.5801
Audio type: hip hop		
		37.2212
Audio type: Pop		
		39.1295
Audio type: jazz		
		40.6835
Audio type: rock		
		38.6835

Fig. 5 Test audio samples used for experimentation in this work

employed for experimental evaluation, complete extraction of the watermark occurs without the application of any attack, and the watermarked audio samples and the host audio samples are indistinguishable from one another. Table 2 compares the proposed watermarking framework to the recent cutting-edge methods. This table shows that the proposed framework performs better than the methods found in the literature.

In addition to this, assessments of the watermarking techniques' subjective quality are essential to assess inaudibility using human perception. It entails getting feedback and preferences from real viewers or listeners who judge how good they think the content is. Particularly in applications like audio watermarking, this kind of assessment is crucial for comprehending how people experience and evaluate the fidelity, clarity, and general quality of multimedia signals. Therefore, an informal subjective listening test was employed in this paper to gauge participants' perceptions of the watermarked audio's quality. During the hearing tests, 100 participants comprising 50 females and 50 males were asked to assess the differences. We have compared host and watermarked audio and employed a Mean Opinion Score

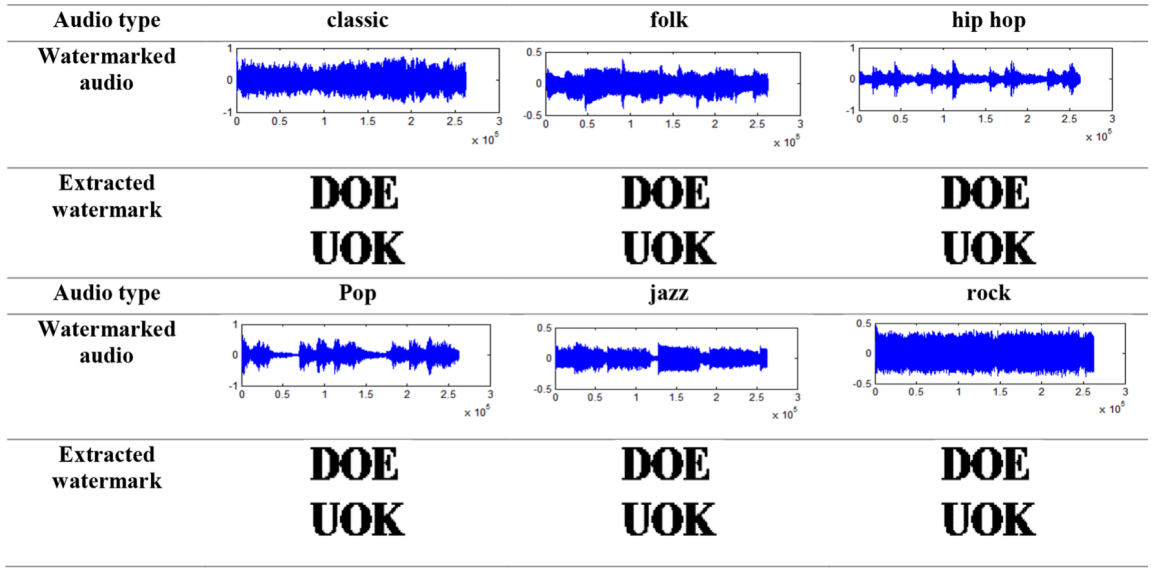


Fig. 6 Watermarked audio samples and their extracted watermarks without any attack

Table 2 Comparison of SNR with contemporary state of the art techniques

Schemes	SNR
Watermarked Audio (Zhang et al., 2023)	25.010
Watermarked Audio (Naqash et al., 2023)	40.050
Watermarked Audio (Wu et al., 2020)	33.060
Watermarked Audio (Liu et.al., 2019)	33.600
Watermarked Audio (Charfeddine et.al., 2022)	40.00
Watermarked Audio (Salah et al., 2021)	36.00
Watermarked Audio (Yamni et al., 2022)	31.4936
Watermarked Audio (Hu et al., 2023)	18.31
Watermarked Audio (proposed)	39.5801
Watermarked audio (average 59 audio samples)	37.1901

Table 3 Subjective listening analysis of the proposed framework

Test subject (Female)	Level of impact	Test subject (Male)	Level of impact
Ist Candidate	5	1st Candidate	5
2nd Candidate	4	2nd Candidate	4
3rd Candidate	5	3rd Candidate	5
4hCandidate	4	4thCandidate	4
:		:	
:		:	
:		:	
:		:	
50thCandidate	5	50thCandidate	4
Mean Score	4.7	Mean Score	4.6

(MOS) of 5. Both observers and hearers rated the sounds on a 5-point scale, where 5 represented highly imperceptible, 4 represented noticeable but not unpleasant, 3 represented somewhat uncomfortable, 2 represented tiresome, and 1 represented extremely annoying. The MOS for the 100 applicants (50 men and 50- women) is displayed in Table 3.

5.2 Robustness evaluation

Robustness is the most extensive measure of copyright protection because it measures how well the watermarking system can endure a variety of attacks. To evaluate robustness, we used a variety of standard signal processing, desynchronization, and stirmark operations to attack the watermarked signal. By performing these actions, the watermark may be eliminated, making it more challenging for the owners of the copyright to identify whether or not the audio signal has the particular watermark. We estimated the NCC and BER as measures of the total resilience of the presented watermarking framework. The attacks subjugated incorporate standard signal processing operations like "Gaussian noise," "low-pass filtering, and "median filtering" as displayed in Table 4, and desynchronization and Stirmark attacks displayed in Table 5 respectively. After extracting and decrypting the watermark, the objective parameters of NCC and BER were assessed to determine the resilience of the provided robust watermarking technique. The robustness test results for typical signal processing, desynchronization, and stirmark attacks are shown visually in Figs. 7, 8, and 9 respectively. This helps the viewer understand the significance of a higher extracted watermark. These numbers show that even after being subjected to various sorts of attacks, the recovered watermarks can still be recognized as NCC is close to 1 in the majority of cases.

We took into consideration two more attacks—desynchronization and stirmark attacks—that are more difficult to carry out to further examine the resilience of the proposed

Table 4 Performance analysis of proposed framework for standard signal processing attacks (for 64 × 64 sized watermark)

Attack Type	Folk_02	Hip hop_10	Average 58 audio samples
	(NCC)		
Gaussian noise (0.001)	1	1	0.9987
Gaussian noise 20 dB	1	1	0.9999
Gaussian noise 10 dB	0.9998	0.9986	0.9985
Gaussian noise 5 dB	0.9630	0.9721	0.9598
Gaussian noise 2 dB	0.9173	0.9301	9202
LPF (fc = 5 K)	1	1	0.9995
LPF (fc = 8 K)	0.9998	0.9995	0.9995
Resampling (1:4)	0.9998	0.9995	0.9997
Resampling (1:8)	0.8979	0.89970	0.9972
Echo adding	0.9970	0.9971	0.9899
Amplitude Scaling (2)	1	1	0.9995
Amplitude scaling (0.5)	1	1	0.9997
Requantization (16bits)	1	0.9998	0.9997
Requantization (24 bits)	1	1	0.9991
HPF (fc = 5 K)	0.9966	0.9964	0.9950
HPF (fc = 8 K)	0.9766	0.9794	0.9690
Median filtering	0.9812	0.9805	0.9802

Table 5 Evaluating the Performance of Suggested Method for Desynchronization and Stirmark Attacks (for 64 × 64 sized watermark)

Attack Type	Folk_02	Hip hop_10	Average 58 audio samples
	(NCC)		
LSB Reset	1	1	0.9999
Cropping (10%)	1	1	0.9997
Cropping (20%)	1	0.9997	0.9999
Cropping (30%)	0.9749	0.9700	0.9699
Jittering (1/1000)	0.9980	0.9982	0.9977
Jittering (1/100)	0.9974	0.9973	0.9978
Jittering (1/10)	0.9956	0.9950	0.9949
Time Scaling (90%)	1	1	0.9998
Time Scaling (110%)	1	0.9998	0.9996
Normalize	1	1	0.9999
Smoothening	1	1	0.9997
Invert	1	1	0.9999
Exchange	0.9934	0.9892	0.9911

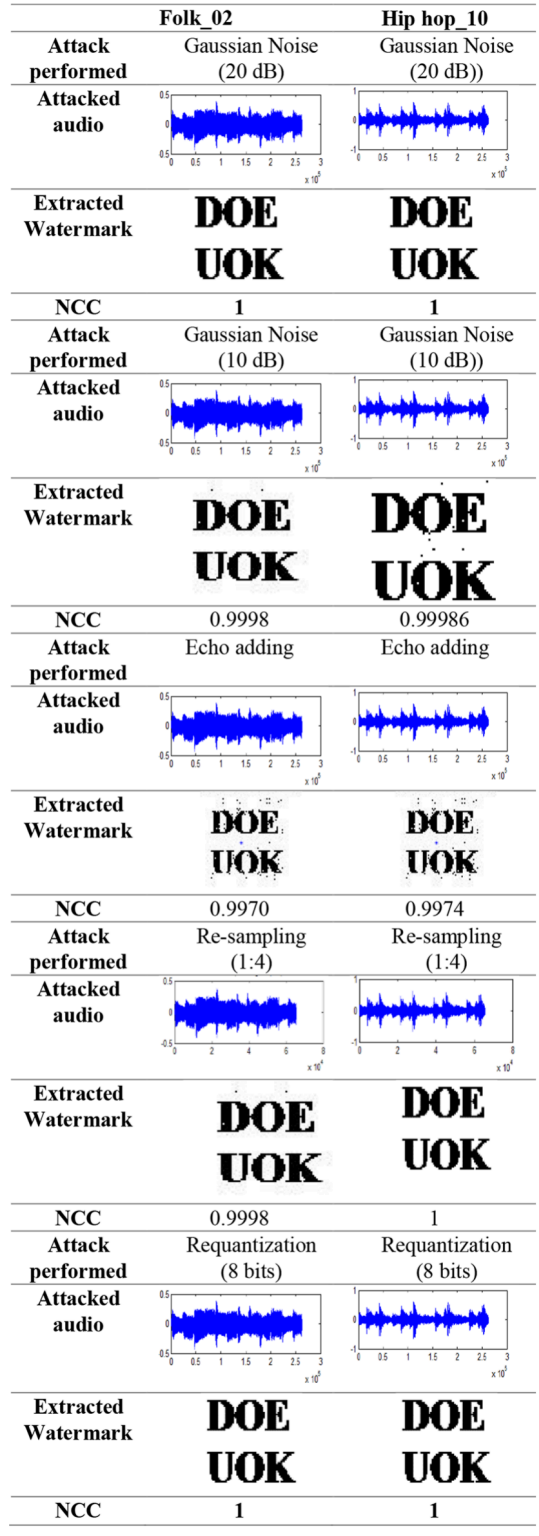


Fig. 7 Watermarked audio and extracted watermarks under common signal processing attacks

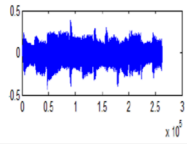
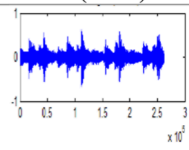
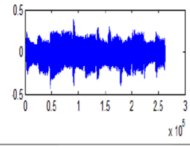
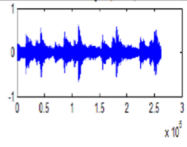
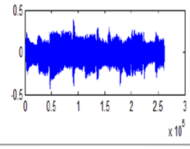
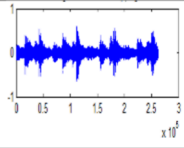
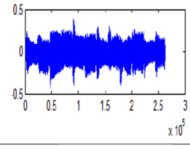
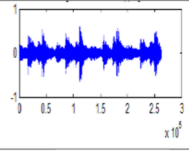
	Folk_02	Hip hop_10
Attack performed	Jittering (1/100)	Jittering (1/100)
Attacked Audio		
Extracted Watermark		
NCC	**0.9974**	**0.9973**
Attack performed	Jittering (1/1000)	Jittering (1/1000)
Attacked audio		
Extracted Watermark		
NCC	**0.9980**	**0.9982**
Attack performed	Cropping (10%)	Cropping (10%)
Attacked audio		
Extracted Watermark		
NCC	1	1
Attack performed	Time scaling (80%)	Time scaling (80%)
Attacked audio		
Extracted Watermark		
NCC	**1**	**1**

Fig. 8 Watermarked audio files along with retrieved watermarks using several desynchronization attacks

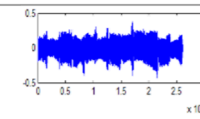
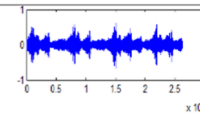
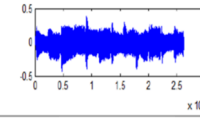
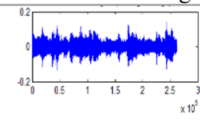
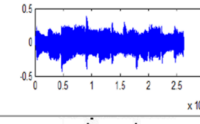
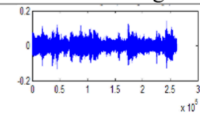
	Folk_02	Hip hop_10
Attacks	Inverse	Inverse
Attacked audio		
Extracted Watermark		
NCC	**1**	**1**
Attacks	Smoothening	smoothening
Attacked audio		
Extracted Watermark		
NCC	**1**	**0.9997**
Attacks	Exchange	Exchange
Attacked audio		
Extracted Watermark		
NCC	**0.9934**	**0.9892**

Fig. 9 Watermarked audio and extracted Watermarks under different stirmark attacks

watermarking method. Deliberate actions that disrupt the synchronization or temporal alignment of different audio channels or constituents are known as desynchronization attacks in audio transmissions. Attackers of this kind aim to manipulate or misalign the time of audio signals, which can result in poor audio quality, changes in perception, or coherence loss.

On the other hand, Stirmark attacks are benchmark attacks that assess the resilience of audio watermarking systems under stress (Zhao et al., 2021). This tests how well the watermark holds up when the audio data is processed in ways that alter its frequency composition. The performance analysis of the proposed algorithm for such kinds of attacks incorporating "cropping", "jittering", "smoothing" etc. in terms of NCC are displayed in Table 5.

Tables 6(a), (b), and (c) shows the comparison of the proposed framework with state-of-the-art techniques in terms of BER. As indicated in Sect. I, our novel blind, secure, and robust audio watermarking technique, which is a purely data-driven-based approach, is immune to many kinds of attacks due to the embedding of the watermark in the residue component. It is clear from these tables that our method performs better than the most advanced schemes.

Table 6(a) Comparative evaluation of presented scheme with various state-of-art

Schemes	(Zhao et al., 2021)	(Zhang et al., 2023)	(Wu et al., 2020)	(Liu et al., 2019)	This work
Attack Type		**BER**			
LPF (fc = 8 K)	0	0	0.050	0.010	0
Requant. (8 bits)	2.69	0.45	0	NA	0
Echo Adding	2.73	2.71	0	2	1.099
Amplitude Scaling(2)	0	0	0.910	0	0
Jittering (1/1000)	1.86	0.340	2.450	0	0.170
Jittering (1/100)	4.880	0.370	NA	1.001	0.229
Jittering (1/10)	12.58	0.480	NA	1.000	0.756

Table 6(b) comparative evaluation of the presented technique with several state-of-art schemes

	(Yamni et al., 2022)	(Naqash et al., 2023)	Proposed
Attack Type	**BER**		
Gaussian noise (15 dB)	0.1342	0.001	0
Re-quantization (8 bits)	0	0	0
Resampling (1:8)	0	0	0.0007
Gaussian noise (20 dB)	0	0.136	0
Cropping (10%)	0	NA	0

Table 6(c) Comparative investigation of the Presented method with different State-of-art Schemes

	(Salah et al., 2021)	(Yamni et al., 2022)	(Charfeddine et al., 2022)	(Xu et al., 2021)	Our work
Attack Type	**NCC**				
LPF (8 K)	0.944	0.987	0.980	0.991	0.999
Requant. (8 bits)	0.961	0.983	1	NA	1
Resampling (1:4)	0.987	0.826	0.869	0.981	0.999
AWGN (20 dB)	0.910	0.892	NA	0.994	1

In Figs. 7 and 8, respectively, the watermarked audio and their derived watermarks under different desynchronization and stirmark attacks are shown visually. These numbers show that after applying a variety of desynchronization and stirmark attacks, watermarks have been fully extracted.

The robustness of the suggested scheme has been analyzed and compared with various techniques available in the literature. It is clear from Table 4 (c) that our scheme performs excellent than the existing schemes. Furthermore, the comparison obtained in Fig. 10 and 11 reveals that our scheme is superior to the state-of-the-art.

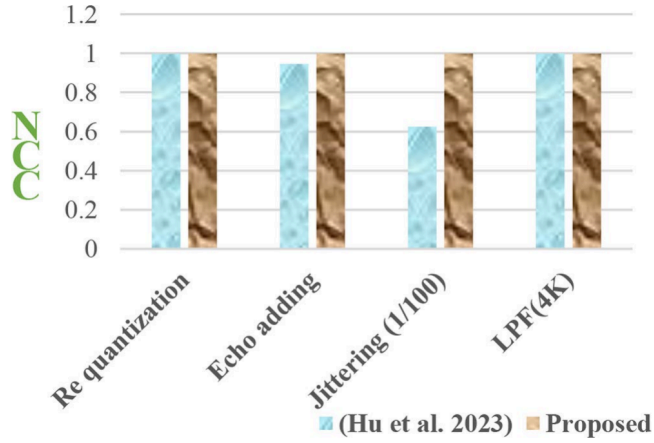


Fig. 10 Comparative investigation of the given technique with (Hu et al., 2023) under different attacks

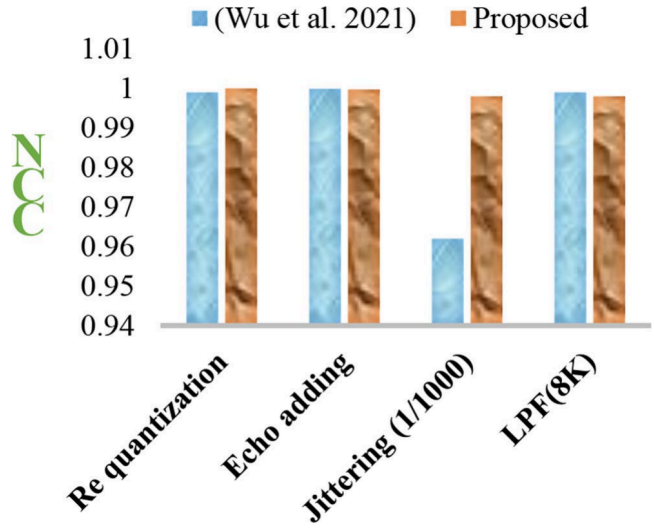


Fig. 11 Comparative investigation of the given technique with (Wu et al., 2020) under different attacks

Embedding capacity or Payload is one of the important parameters of any watermarking framework. Figure 12 reveals that the proposed scheme outperforms the schemes available in the literature.

To further examine and enhance the embedding capacity of the proposed watermarking system, we have increased the watermark size from 64×64 to 128×128. This modification significantly improves the payload (from 204.8 bps to 819.2 bps) without compromising the robustness or imperceptibility of the watermarked audio. The performance of the system with the larger watermark is presented in Tables 7 and 8 in terms of common signal processing, desynchronization, and stirmark attacks. Even under the increased payload capacity, the system exhibits high robustness and retains high perceptual quality with SNR measures of 36.024 dB for folk and 34.548 dB for hip-hop audio loops. The findings validate that the suggested arrangement achieves a well-balanced

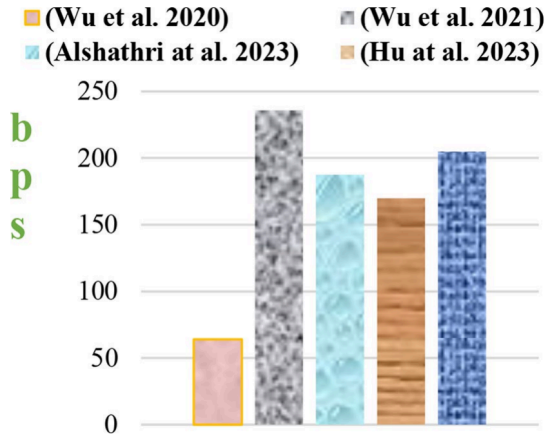


Fig. 12 Payload evaluation of presented scheme with multiple state-of-the-art

Table 7 Performance analysis of proposed framework for standard signal processing attacks (for 128 × 128 sized watermark)

Attack Type	Folk_02 (NCC)	Hip hop_10
Gaussian Noise (0.001)	0.9993	0.9987
Gaussian Noise 20 dB	0.9991	0.9987
Gaussian Noise 10 dB	0.9974	0.9960
Gaussian Noise 5 dB	0.9511	0.9622
Gaussian Noise 2 dB	0.8859	0.9021
LPF (fc = 5 K)	0.9987	0.9982
LPF (fc = 8 K)	0.9983	0.9976
Resampling (1:4)	0.9976	0.9969
Resampling (1:8)	0.8724	0.8781
Echo adding	0.9954	0.9946
Amplitude Scaling (2)	0.9971	0.9964
Amplitude scaling (0.5)	0.9960	0.9951
Requantization (16bits)	0.9964	0.9956
Requantization (24 bits)	0.9873	0.9842
HPF (fc = 5 K)	0.9910	0.9893
HPF (fc = 8 K)	0.9882	0.9870
Median filtering	0.9731	0.9701

solution between embedding capacity, robustness, and imperceptibility, and hence a good candidate for real-world high-payload watermarking applications.

5.3 Timing analysis

An important component of evaluating the efficiency and speed of the proposed watermarking system is timing analysis. In this analysis, the time needed for essential tasks, notably the watermark embedding and the subsequent watermark extraction using iterative filtering together with adaptive embedding is measured. Table 9 will display a thorough investigation of the times involved in both the embedding and extraction stages and watermark encryption and decryption times. This analysis is crucial for evaluating

Table 8 Performance analysis of proposed framework for desynchronization and stirmark attacks (for 128 × 128 sized watermark)

Attack Type	Folk_02 (NCC)	Hip hop_10
LSB Reset	0.9996	0.9995
Cropping (10%)	0.9993	0.9991
Cropping (20%)	0.9994	0.9971
Cropping (30%)	0.9599	0.9608
Jittering (1/1000)	0.9959	0.9962
Jittering (1/100)	0.9957	0.9964
Jittering (1/10)	0.9936	0.9939
Time Scaling (90%)	0.9996	0.9992
Time Scaling (110%)	0.9992	0.9991
Normalize	0.9998	0.9995
Smoothening	0.9994	0.9990
Invert	0.9992	0.9995
Exchange	0.9915	0.9877

Table 9 Timing investigation of test audio samples (in Seconds)

Algorithm used	Clock frequency = 3 GHz	
	Encryption time (sec)	Decryption time (sec)
Chaotic encryption	0.0782	0.0001
DNA Encryption	0.7812	0.0002
Audio loop	embedding time (sec)	Extraction time (sec)
Folk_02	1.9625	1.7193
Hip hop_10	1.9571	1.7181

the computational complexity of our suggested watermarking framework.

One limitation of the given watermarking scheme is its relatively high embedding and extraction times, as displayed in Table 9, which exceed the typical threshold for hard real-time audio processing (generally under 10 milliseconds). While the current performance is sufficient for offline applications including content authentication or copyright protection in post-production environments where little delays are acceptable, it is not yet suitable for real-time deployment. In future work, we aim to address this limitation by enhancing the computational efficiency of the system to enable its applicability in hard real-time scenarios.

6 Conclusion

This study presents a new audio watermarking methodology based on embedding in intrinsic modes that addresses the challenges of digital piracy and copyright infringement in the modern era. Our approach, based on iterative filtering and enhanced by double-layer encryption using chaos and DNA coding, offers a robust, secure, and blind solution for audio content protection. The proposed method demonstrates excellent performance in terms of imperceptibility, payload capacity, and robustness, achieving perfect NCC

values under no-attack conditions. Unlike many existing techniques, our method shows remarkable resilience against not only common signal processing attacks but also desynchronization and stirmark attacks. The blind nature of the system, coupled with its superior performance in balancing imperceptibility, payload capacity, and robustness, makes it well-suited for protecting audio transmissions and intellectual property in real-world scenarios. This work represents a significant advancement in audio watermarking technology, providing a versatile solution for copyright protection and content authentication in the digital age, with potential applications across various multimedia security domains. However, the computational complexity may limit real-time applications. Future research can explore advanced transformations, and computational optimizations for large-scale deployments to further strengthen robustness, security, and computational complexity in particular. Overall, this framework offers a promising approach for robust audio copyright protection and content authentication.

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Declarations

Competing interests The authors declare no competing interests.

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